

1.0 INTRODUCTION AND OBJECTIVES

The Olist E-Commerce dataset contains information on customer purchases, product categories, payment methods, seller locations, customer reviews, marketing lead acquisition, and closed deals. The dataset provides a comprehensive view of the e-commerce business lifecycle, from customer acquisition to order fulfillment and customer satisfaction.

The primary objective of this analysis is to evaluate Olist's sales performance, customer behavior, seller distribution, marketing effectiveness, and operational efficiency. Through Power BI, interactive dashboards were developed to identify trends, measure key performance indicators (KPIs), and generate actionable business insights.

The analysis specifically aims to:

- Evaluate overall sales and order performance.
- Identify top-performing product categories.
- Analyze customer satisfaction trends.
- Assess the effectiveness of marketing acquisition channels.
- Examine payment preferences and spending behavior.
- Understand geographic distributions of customers and sellers.
- Investigate operational factors affecting customer satisfaction.
- Provide recommendations for improving revenue growth, customer experience, and operational performance.

2.0 METHODOLOGY

2.1 Data Collection

The analysis utilized multiple Olist datasets including orders, customers, sellers, products, payments, reviews, marketing qualified leads, closed deals, and product category translations.

2.2 Data Preprocessing

Several preprocessing steps were performed to improve data quality and usability:

- Imported and combined multiple datasets using Power BI relationships.
- Created a centralized Date Table to support time-based analysis and slicers.
- Established relationships between Orders, Customers, Products, Payments, Reviews, Sellers, and Marketing datasets.
- Translated product category names from Portuguese to English using the Product Category Translation table.
- Replaced missing category values with "Unknown" where appropriate.
- Converted date and datetime fields into proper Power BI date formats.
- Created calculated measures and KPIs using DAX.
- Removed or handled blank and null values that could affect analysis accuracy.
- Created active customer and active seller measures to ensure time-based filtering functionality.

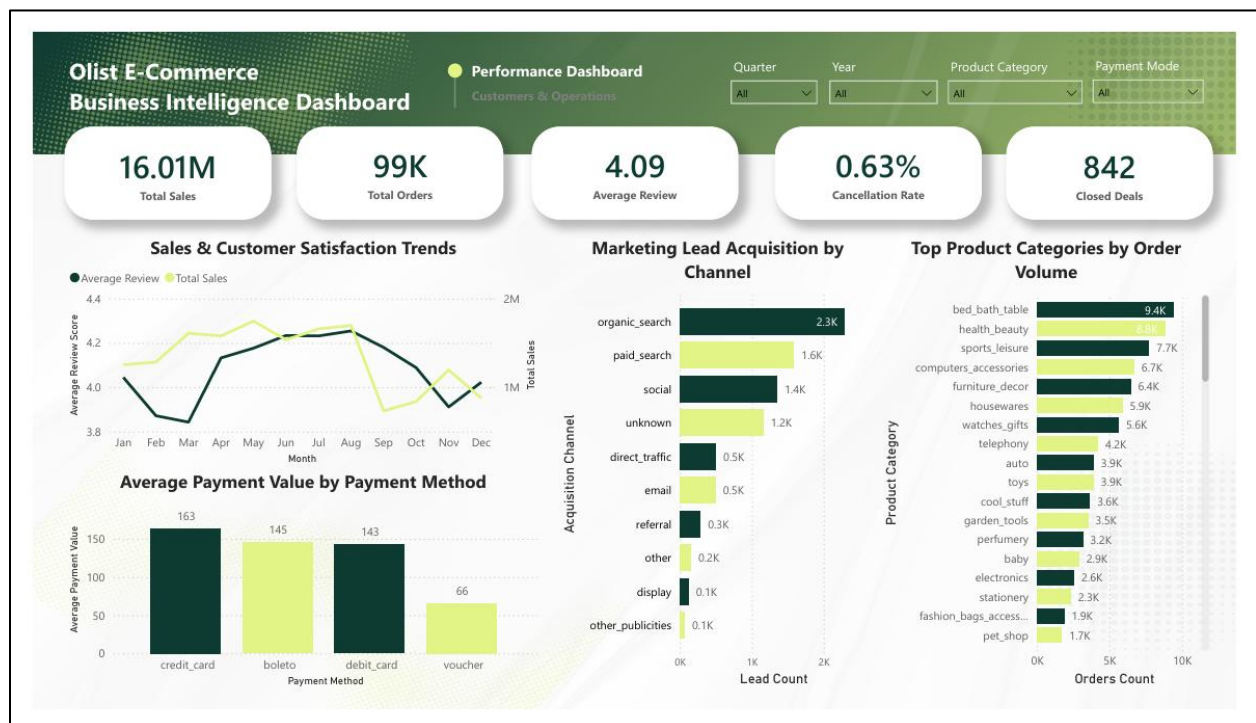
2.3 Analysis Techniques

The following metrics were calculated using DAX:

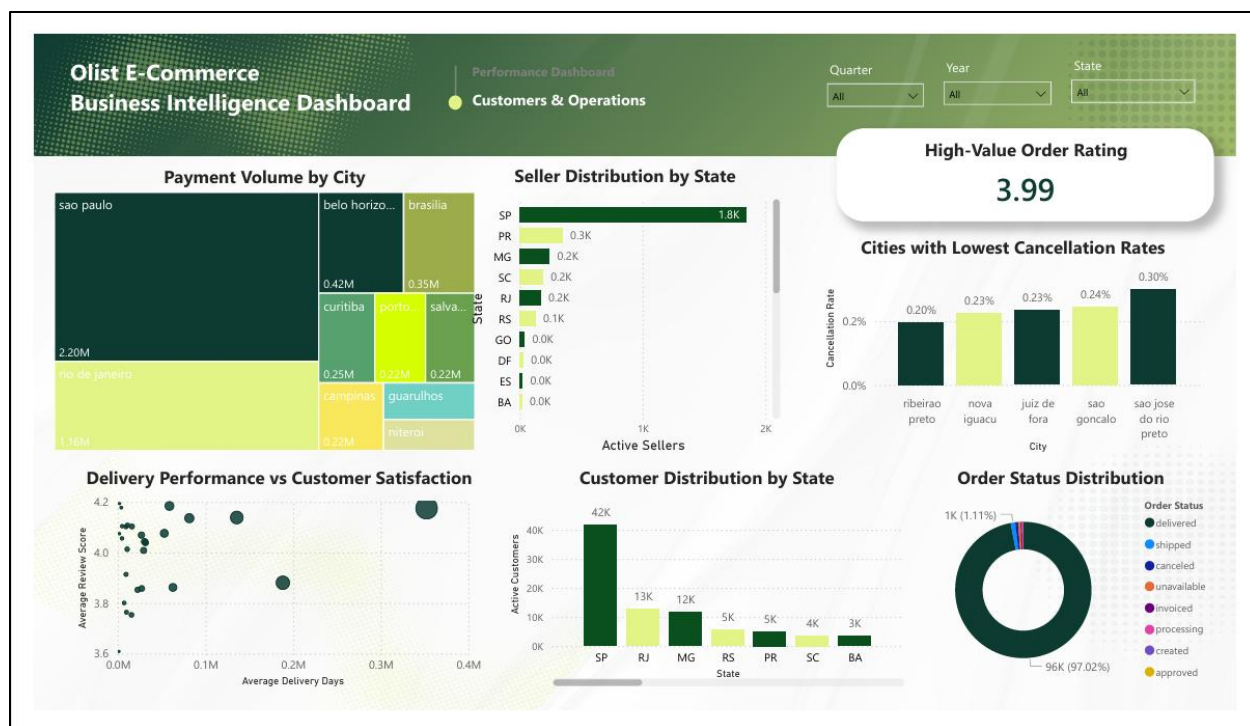
- Total Sales
- Total Orders
- Average Review Score
- Cancellation Rate
- Closed Deals Count
- Active Customers
- Active Sellers
- Average Payment Value
- High-Value Order Rating
- Delivery Performance Metrics

3.0 RESULTS

To effectively analyze the Olist E-Commerce dataset, a comprehensive two-page interactive Business Intelligence dashboard was developed using Microsoft Power BI. The dashboard was designed to transform raw transactional, customer, seller, marketing, and operational data into meaningful business insights through a combination of key performance indicators (KPIs), charts, maps, slicers, and interactive visualizations.



The first dashboard page focuses on Sales and Business Performance, providing an overview of revenue generation, customer satisfaction trends, product category performance, payment behavior, and marketing effectiveness. Interactive filters allow users to analyze performance across different periods, product categories, and payment methods, enabling deeper exploration of sales trends and customer purchasing behavior.



The second dashboard page focuses on Customers and Operations, highlighting customer and seller distribution, delivery performance, order fulfillment status, cancellation rates, and geographic business concentration. Through visualizations such as treemaps, scatter plots, bar charts, and donut charts, the dashboard provides a comprehensive view of operational efficiency and customer experience across different regions.

Together, these dashboards enable stakeholders to monitor key business metrics, identify trends and opportunities, evaluate operational performance, and support data-driven decision-making. The following sections present the critical insights derived from the analysis and discuss their business implications.

3.1 Overall Business Performance



The analysis revealed that Olist generated approximately 16.01 million in total sales from 99,000 orders, while maintaining an average customer review score of 4.09 out of 5 and a relatively low cancellation rate of 0.63%. These figures indicate a healthy e-commerce marketplace with strong customer engagement and generally positive customer experiences. Additionally, 842 closed deals demonstrate successful seller acquisition and marketplace growth initiatives.

Business Implication

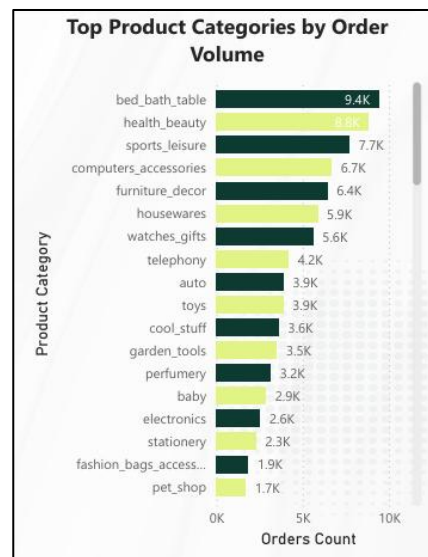
The combination of high sales volume, strong customer satisfaction, and low cancellation rates suggests that Olist has established an effective platform capable of supporting both customers and sellers while maintaining operational reliability.

3.2 Product Category Performance

The analysis of order volume shows that demand is concentrated among a small number of product categories. The leading categories include:

- Bed Bath & Table
- Health & Beauty
- Sports & Leisure
- Computers & Accessories
- Furniture & Decor

Together, these categories account for a substantial portion of marketplace transactions.



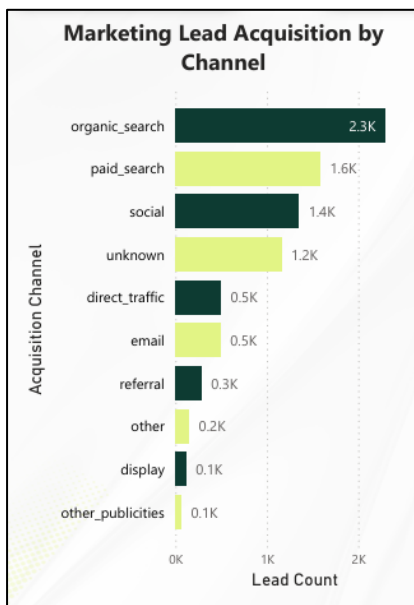
Business Implication

The concentration of demand indicates that Olist's revenue is heavily influenced by a few key categories. Inventory planning, supplier partnerships, and marketing investments should prioritize these categories to maximize revenue generation while maintaining adequate stock levels.

Recommendation

Develop targeted campaigns around top-performing categories and investigate opportunities to improve underperforming categories to diversify revenue streams.

3.3 Marketing Channel Effectiveness



Organic Search emerged as the strongest acquisition channel, generating approximately 2.3 thousand leads, followed by Paid Search and Social channels.

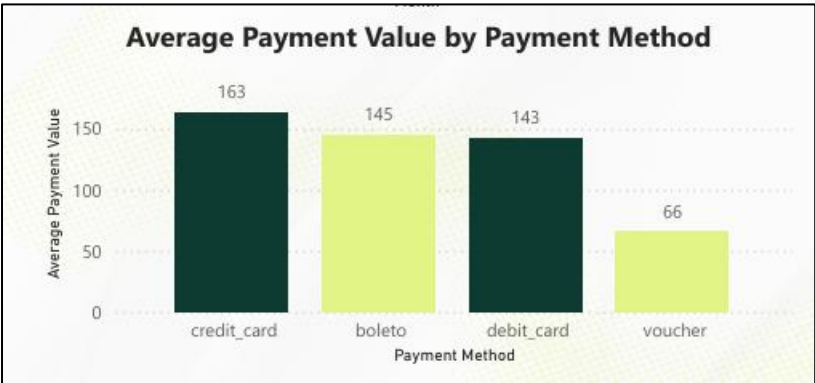
Business Implication

Organic Search represents the most efficient source of new leads, suggesting that customers actively discover Olist through search engines and online content.

Recommendation

Increase investments in SEO, content marketing, and search visibility strategies to capitalize on this highly effective acquisition channel while optimizing spending on lower-performing channels.

3.4 Payment Behavior Analysis



Credit card transactions produced the highest average payment value, significantly outperforming vouchers and slightly exceeding debit card and boleto payments.

Business Implication

Customers using credit cards tend to place larger orders, potentially due to installment options and greater purchasing flexibility.

Recommendation

Introduce payment incentives, installment promotions, and loyalty programs that encourage higher-value credit card purchases.

3.5 Sales and Customer Satisfaction Trends



The analysis of monthly sales performance and customer satisfaction reveals that Olist maintained consistently high review scores throughout the year despite fluctuations in sales volume. Average review scores remained close to or above 4.0 across most months, indicating a generally positive customer experience.

Sales performance peaked between May and August, reaching nearly 2 million in monthly sales, before experiencing a noticeable decline during September and October. Despite these fluctuations, customer satisfaction remained relatively stable and continued to exceed 4.0 during peak sales periods.

A notable observation is that higher sales volumes did not result in a decline in customer satisfaction. This suggests that Olist's operational processes, seller network, and fulfillment capabilities were able to accommodate increased demand without significantly affecting the customer experience.

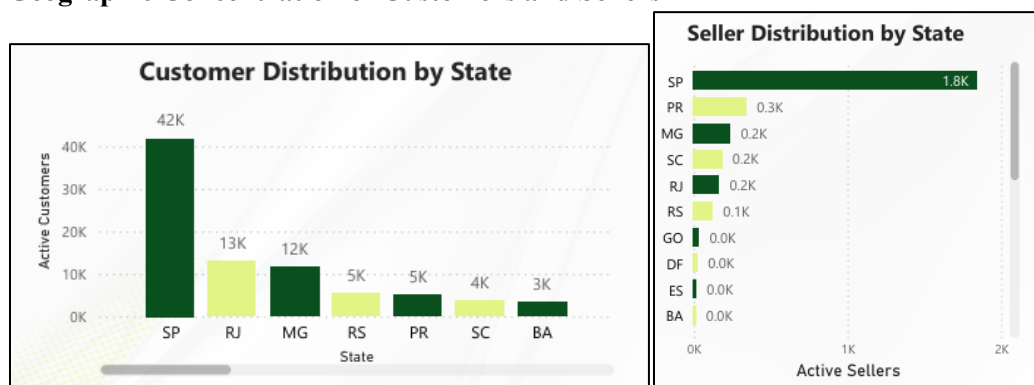
Business Implication

The ability to maintain strong customer satisfaction during periods of high sales activity indicates operational scalability and effective marketplace management. This is an important indicator of business maturity, as many e-commerce platforms experience declining service quality during peak demand periods.

Recommendation

Olist should further analyze the factors contributing to the sales peak observed between May and August, including seasonal demand, promotional campaigns, and product category performance. Replicating these successful strategies while maintaining operational efficiency may help sustain revenue growth throughout the year. Additionally, maintaining the operational practices that support consistently high review scores should remain a strategic priority to preserve customer loyalty and repeat purchases.

3.6 Geographic Concentration of Customers and Sellers



The analysis shows that both customers and sellers are highly concentrated within a small number of Brazilian states.

Customers

- Sao Paulo (SP): 41,476 active customers
- Rio de Janeiro (RJ): 12,852 active customers
- Minas Gerais (MG): 11,635 active customers

Sellers

- Sao Paulo (SP): 1,849 active sellers
- Paraná (PR): 349 active sellers
- Minas Gerais (MG): 244 active sellers

Business Implication

The marketplace demonstrates significant dependence on Sao Paulo as both its primary customer market and seller hub. While this concentration creates operational efficiencies, it also introduces geographic risk.

Recommendation

Expand seller recruitment and customer acquisition campaigns in underrepresented states to diversify revenue sources and reduce operational dependence on a single region.

3.7 Delivery Performance and Customer Satisfaction



The Delivery Performance vs Customer Satisfaction analysis indicates a relationship between delivery efficiency and customer reviews.

States with shorter delivery durations generally achieve higher average review scores, while longer delivery periods are associated with lower satisfaction levels.

Business Implication

Delivery performance remains one of the most influential factors affecting customer perception and platform satisfaction.

Recommendation

Continue investing in logistics optimization, warehouse management, and delivery tracking systems to improve delivery speed and customer experience.

3.8 Cancellation Rate Analysis

Several cities consistently demonstrate exceptionally low cancellation rates:

- Ribeirão Preto
- Nova Iguaçu
- Juiz de Fora
- São Gonçalo
- São José do Rio Preto

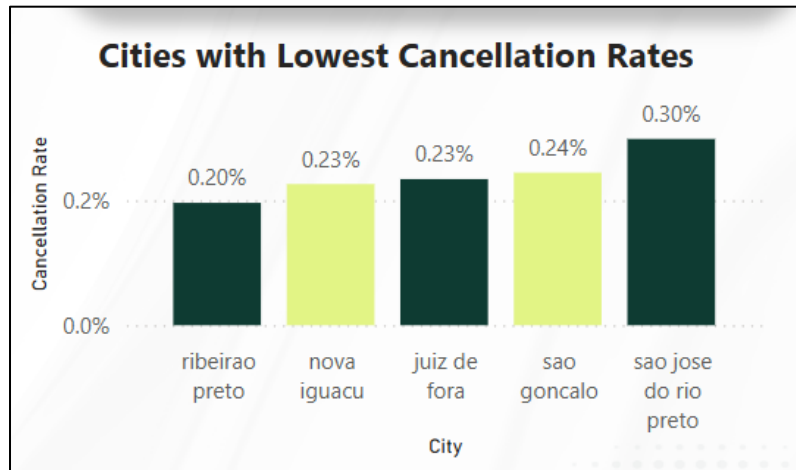
All maintain cancellation rates below approximately 0.30%.

Business Implication

These locations represent operational benchmarks that may provide insights into effective fulfillment practices, customer communication strategies, and logistics performance.

Recommendation

Conduct further investigation into the operational characteristics of these cities and replicate successful practices across other regions.

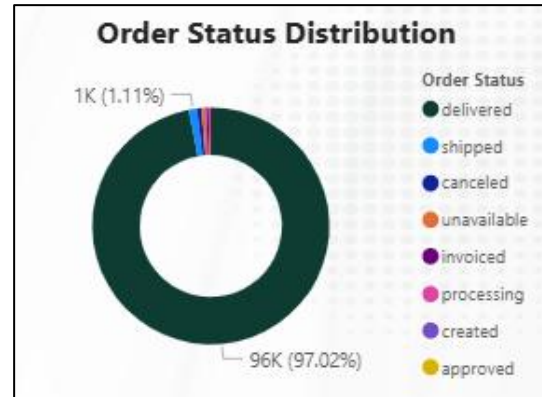


3.8 Order Fulfillment Performance

The overwhelming majority of orders were successfully delivered, accounting for approximately 97% of all orders, while canceled, unavailable, invoiced, processing, and approved orders collectively represent only a small fraction of transactions.

Business Implication

This indicates that Olist maintains highly effective order fulfillment operations and demonstrates strong reliability in delivering purchased products to customers.



The low proportion of canceled and unavailable orders suggests that inventory management and fulfillment processes are generally functioning efficiently.

Recommendation

Maintain current fulfillment standards while focusing on reducing the remaining order exceptions through improved inventory forecasting, supplier coordination, and proactive customer communication.

4.0 RECOMMENDATIONS

Based on the analysis conducted, several strategic opportunities have been identified to improve Olist's operational performance, customer satisfaction, and long-term business growth.

4.1 Strengthen Organic Marketing Efforts

Organic Search was identified as the most effective lead acquisition channel, generating the highest number of marketing-qualified leads compared to other channels. Olist should continue investing in search engine optimization (SEO), content marketing, and website visibility initiatives to increase customer acquisition while maintaining cost efficiency. Improving search rankings and expanding content coverage can help attract more qualified leads and reduce dependence on paid advertising.

4.2 Expand Seller Recruitment Across Underrepresented Regions

The analysis revealed a significant concentration of sellers in Sao Paulo, while many regions maintain relatively low seller representation. Expanding seller recruitment programs in underrepresented states can improve product availability, reduce logistical bottlenecks, and strengthen marketplace resilience. A more geographically balanced seller network may also contribute to faster deliveries and improved customer satisfaction.

4.3 Prioritize High-Performing Product Categories

Product categories such as Bed Bath & Table, Health & Beauty, Sports & Leisure, Computers & Accessories, and Furniture & Decor consistently generated the highest order volumes. Olist should prioritize inventory management, supplier partnerships, promotional campaigns, and category-specific marketing initiatives within these high-performing segments to maximize revenue opportunities and maintain customer demand.

4.4 Improve Delivery Performance and Logistics Efficiency

The analysis demonstrated a relationship between delivery performance and customer satisfaction. Faster delivery times were generally associated with higher review scores, indicating that logistics remains a critical factor influencing customer experience. Olist should continue investing in warehouse optimization, delivery routing, inventory forecasting, and logistics partnerships to further reduce delivery times and improve customer satisfaction.

4.5 Encourage Higher-Value Transactions

Credit card payments generated the highest average payment values among all payment methods. To increase average order value, Olist may introduce installment-based promotions, loyalty incentives, cashback programs, and strategic partnerships with payment providers. Encouraging customers to utilize payment methods associated with higher spending can positively impact overall revenue generation.

4.6 Diversify Customer Acquisition Across Regions

The customer base is highly concentrated in a small number of states, particularly Sao Paulo, Rio de Janeiro, and Minas Gerais. While these regions represent strong markets, geographic concentration creates business risk. Olist should develop targeted regional marketing campaigns and localized seller partnerships to expand its presence in emerging markets and reduce dependence on a limited number of locations.

4.7 Areas for Further Study

Future analyses may focus on:

- Customer retention and repeat purchase behavior.
- Customer lifetime value (CLV) analysis.
- Product profitability by category.
- Seasonal demand forecasting.
- Delivery performance by logistics partner.
- Regional profitability and operational efficiency.
- Predictive modeling for cancellations and customer satisfaction.

These studies can provide deeper insights that support long-term strategic planning and operational improvements.

5.0 CONCLUSION

This analysis provided a comprehensive assessment of Olist's sales performance, customer behavior, marketing effectiveness, and operational efficiency through the development of interactive Power BI dashboards.

The findings indicate that Olist maintains strong marketplace performance, generating approximately 16.01 million in sales across 99,000 orders while achieving a high average customer review score and a low cancellation rate. Organic Search emerged as the most effective lead acquisition channel, while Credit Card transactions generated the highest average payment values. Additionally, several product categories demonstrated consistently strong demand, highlighting key revenue drivers within the platform.

From an operational perspective, delivery performance was found to be closely associated with customer satisfaction, reinforcing the importance of logistics efficiency. The analysis also revealed significant geographic concentration among both customers and sellers, particularly within Sao Paulo, presenting opportunities for regional expansion and risk diversification.

Overall, the results suggest that Olist is operating as a healthy and efficient e-commerce marketplace with strong customer satisfaction and fulfillment performance. By strengthening organic marketing efforts, expanding seller recruitment across underserved regions, improving logistics operations, and focusing on high-performing product categories, Olist can further enhance customer experience, increase revenue potential, and support sustainable long-term growth.

The dashboard and insights generated through this analysis provide a data-driven foundation for informed decision-making and future business optimization initiatives.